



Changmin Lee

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| 🎓 Changmin Lee | 🏢 UAI Lab, UNIST

Education

Ulsan National Institute of Science and Technology (UNIST)

Ulsan, Korea

M.S.-PH.D. INTEGRATED PROGRAM, COMPUTER SCIENCE AND ENGINEERING

Mar. 2025 – Present

- UAI Lab, Advisor: Prof. Taesik Gong
- Research focus: Personalized AI, on-device RAG, efficient LLMs, and AI agents.

Kyungpook National University

Daegu, Korea

B.S., COMPUTER SCIENCE AND ENGINEERING

Mar. 2019 – Feb. 2025

- Total GPA: **3.9/4.3** Major GPA: **4.2/4.3**

Research Interests

- **Personalized AI & RAG** (user-specific memory construction, preference-aligned retrieval)
- **on-device AI** (efficient LLM inference under tight resource budgets)
- **AI agents & memory** (compact memory for GUI agents, long-horizon task planning)

Research Experience

UAI Lab, UNIST

Ulsan, Korea

GRADUATE RESEARCHER

Mar. 2025 – Present

- Conduct research on personalized AI, on-device RAG, efficient memory systems, and resource-efficient AI agents.
- Designed EPIC, a preference-aligned memory construction and retrieval framework for efficient personalized RAG under on-device resource constraints. EPIC improved preference-following accuracy by 18.79 percentage points on average across four benchmarks while reducing memory usage by 2,404× and retrieval latency by 32.17× compared with best performed baseline; accepted at **ICML 2026** and formed the basis of a Korean patent application.
- Investigate transition-guided memory and flexible action reuse for efficient mobile GUI agents, aiming to reduce redundant model inference during task execution.

Pattern Recognition & Machine Intelligence Lab, Kyungpook National University

Daegu, Korea

UNDERGRADUATE RESEARCH ASSISTANT

Dec. 2023 – Sep. 2024

- Conducted research on explainable AI, computer vision, medical image analysis, and machine learning.
- Developed a hybrid Transformer-CNN model for chest X-ray classification.
- Participated in regular research paper seminars and presented recent papers in computer vision and medical AI.

Publications

- **“From Volume to Value: Preference-Aligned Memory Construction for On-Device RAG,”**
Changmin Lee, Jaemin Kim, and Taesik Gong,
International Conference on Machine Learning (ICML), 2026.
- **“Enhancing Chest X-Ray Classification with Multi-Class Token Hybrid Transformers,”**
Changmin Lee, Ho-kyung Shin, and Woo-Jeoung Nam,
Journal of The Institute of Electronics and Information Engineers, 2024.
- **“Comparison of Recycling Waste Awareness Performance by Transfer Learning Model,”**
Dong-hyuk Kim, Changmin Lee, et al.,
Korean Multimedia Society Conference, 2024.

Manuscripts in Preparation

- “**△Agent: Efficient Mobile GUI Agents via Transition-Guided Flexible Action Reuse,**”
Changmin Lee, Taehwan Park, Jongwon Lee, and Taesik Gong,
2026. Manuscript in preparation.

Patent Applications

- “**Preference-Based Index Management Method for Generating Personalized Search Responses and System Supporting the Same,**”
Taesik Gong, Changmin Lee, and Jaemin Kim,
Republic of Korea Patent Application No. 10-2025-0182629, filed Nov. 26, 2025. Patent pending.

Awards & Fellowships

Aug. 2025 – Sep. 2026	NRF Master’s Fellowship , Graduate Student Research Encouragement Grant, National Research Foundation of Korea
2024	Excellence Award , Undergraduate Research Support Program, Convergent Medical Scientist Training Program

Teaching Experience

Computer Networks (CSE351), UNIST	
TEACHING ASSISTANT	Fall 2025
• Supported course operations, grading, and student communication. • Assisted students with course materials, assignments, and project questions.	
On-Device AI for Smart Manufacturing (NOVA508), UNIST	
TEACHING ASSISTANT	Spring 2026
• Supported project-based activities and hands-on practice in on-device AI deployment. • Assisted students with assignments and implementation-related questions.	
Data Preprocessing and Visualization, AI Novatus Academia & LG Electronics LDC	
TEACHING ASSISTANT	2025 – 2026
• AI Novatus Academia (Gyeongnam): 6th Cohort (May 2025), 7th Cohort (May–Jun. 2026) • AI Novatus Academia (Ulsan): 8th Cohort (Jul. 2025) • LG Electronics LDC Program: Jul. 2025 and Jan. 2026 • Assisted with data cleaning, transformation, exploratory data analysis, and visualization.	
Artificial Intelligence Fundamentals, AI Novatus Academia (Gyeongnam), 7th Cohort	
TEACHING ASSISTANT	May – Jun. 2026
• Supported lectures and student exercises on fundamental AI concepts.	

Industry-Academic Project Mentoring

LG Electronics HS Future Technology Project	
PROJECT MENTOR	Jun. – Sep. 2025
• Mentored a team developing a personalized model based on Mixture-of-Experts (MoE) for user-specific adaptation. • Provided technical guidance on personalization methods, model architecture, experimental design, and evaluation.	
Vacuum Cleaner Personalization Pipeline, LG Electronics DIC Program PBL	
PROJECT MENTOR	Aug. – Sep. 2025
• Mentored a team designing an end-to-end personalization pipeline for LG vacuum cleaner user scenarios.	
Cutting Tool Dynamic Characteristics Prediction, AI Novatus Academia (Gyeongnam), 6th Cohort	
PROJECT MENTOR	Jun. – Sep. 2025
• Mentored a team developing a model to predict dynamic characteristics of machining tools from cutting-process images.	
Strawberry Smart Farm Growth Cycle Prediction, AI Novatus Academia (Ulsan), 8th Cohort	
PROJECT MENTOR	Sep. – Dec. 2025
• Mentored a team developing a growth-cycle classification model for smart-farm strawberry cultivation.	

Leadership & Service

UAI Lab, UNIST

LAB MANAGER

Ulsan, Korea

Jan. – Dec. 2025

- Coordinated lab operations, internal communication, meeting schedules, and administrative tasks.
- Supported onboarding and communication among lab members.

Undergraduate Project Experience

Recycling Waste Sorting Based on Transfer Learning

TEAM MEMBER

Mar. – Jun. 2024

- Industry-university linkage project; labeled 30-category waste dataset from plant video footage.
- Fine-tuned YOLOv9, YOLOv8, and RT-DETR via transfer learning; proposed best-performing YOLOv9 to the company.

Smart Table Clock

TEAM MEMBER

Sep. – Dec. 2023

- Arduino-based smart clock with LCD widgets, Daegu bus API integration, and Wi-Fi web control.

Remote File Explorer

TEAM LEADER

Mar. – Jun. 2023

- Built a terminal-based remote file explorer in C using the curses library with Linux-style commands.
- Enabled file browsing and transfer between macOS and Linux systems over a network.

Skills

Programming	Python, C, Java
Deep Learning	PyTorch, TensorFlow, Keras, scikit-learn
RAG & Retrieval	FAISS, dense retrieval, LLM-based evaluation, personalized RAG
Systems & Tools	Linux, Git, Docker, LaTeX, vLLM